

Spotify Streaming Analysis Report

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Introduction:

This dataset exhibits a collection of metadata of Spotify tracks of various music genres. Metadata such as artist(s) name, album, track title, popularity score, duration, and whether a track is marked as explicit. Additionally, it provides audio attributes such as danceability, energy, acousticness, instrumentality, and tempo therefore characterizing songs beyond surface-level descriptions, which allows for a deeper and more insightful understanding and analysis.

The main goal of this project is to analyze patterns in a song's production, popularity, and traits that would apply to certain genres. Some questions that this analysis attempts to answer are:

- How do various audio features correlate with a track's popularity on Spotify?
- What audio characteristics are most representative of different musical genres?
- What makes a popular song *popular*?

After addressing these questions, this analysis offers insights to the evolution of music streaming. Following this report, it will inform those that are passionate about music about trends, preferences, and other musical statistics that are to be found.

Exploratory Analysis:

Initial Data Overview:

- The initial dataset comprised 114,000 entries and 21 columns.
- The dataset included one Boolean, nine Float, six Integer, and five Object type columns.
- One missing value was identified within the dataset.

Handling Missing Values and Column Information:

- A single row, corresponding to Token 65900, was found to have incomplete fundamental song information, specifically the artist's name, album name, and track name. Consequently, this incomplete row was removed from the dataset.
- Following the removal of the row with the missing value, the dataset now contains 113,999 entries.
- The 'Unnamed: 0' column was determined to be non informative and was therefore excluded from the analysis.
- The dataset initially appeared to contain duplicate entries, which typically necessitate removal to avoid redundancy.

- However, further investigation revealed that these apparent duplicates originated from the 'track_genre' column, where individual songs could be associated with multiple genres. Therefore, these entries were retained to preserve the multi genre representation of the tracks.
- On the contrary, during statistical testing, duplicate tracks will be dropped to prevent the data from being skewed.

Analysis of Numeric Variables and Visualization:

- A correlation analysis was performed on the continuous variables:
 - *danceability*
 - *energy*
 - *acousticness*
 - *tempo*
 - *loudness*
 - *speechiness*
 - *valence*

Insights from the Continuous Variable Heatmap:

- A positive correlation was observed between *valence* (indicating positivity) and *danceability*, suggesting that more positive songs tend to be more danceable.
- *Energy* and *loudness* exhibited a positive relationship, where louder songs generally possess higher energy levels.
- Conversely, *acousticness* showed a negative correlation with *energy*, implying that more acoustic songs tend to have lower energy.
- For the most part, *acousticness* and *loudness* were negatively correlated, suggesting that acoustic tracks are generally quieter.
- *Speechiness*, which measures the presence of spoken words, displayed a weak positive correlation with both *danceability* and *energy*, indicating a slight tendency for songs with spoken elements to be more danceable or energetic, although this relationship is not strong.

Relationship between Average Loudness and Energy Levels:

Note: Duplicate songs were dropped

- A line plot indicates a generally logarithmic-like increase in average loudness as energy levels rise.

Relationship between Average Acousticness and Energy Levels:

Note: Duplicate songs were dropped

- The line plot reveals an inverse relationship, with average acousticness decreasing as energy levels increase.

Relationship between Average Loudness and Acousticness Levels:

Note: Duplicate songs were dropped

- The line plot shows a general decline in loudness with higher acousticness, punctuated by a significant drop in decibels at the higher end of the acousticness spectrum.

Relationship between Average Danceability and Valence Levels:

Note: Duplicate songs were dropped

- The line plot illustrates an upward trend, demonstrating that average danceability increases with higher valence scores.

Relationship between Average Loudness and Tempo Levels:

Note: Duplicate songs were dropped

- The line plot indicates that average loudness generally increases with tempo, exhibits a slight dip, and then concludes with a significant decrease at the highest tempos.

Distribution of Numeric Variables:

- The distribution of danceability appeared relatively uniform, which is expected given the diverse nature of music, encompassing tracks intended for dancing and those for listening.
- The energy distribution showed a general upward trend, suggesting a prevalence of songs with progressively higher energy levels within the dataset.
- The dataset contained a limited number of acoustic songs, as indicated by the distribution.
- The tempo of the songs was widely distributed, reflecting a variety of paces across the dataset.
- The loudness distribution exhibited a bell shaped curve, likely influenced by the characteristics of different music genres.
- The valence distribution was generally high across most counts, indicating a prevalence of songs with positive sentiment.
- The presence of outliers in the visualizations and distributions did not appear to significantly skew the overall data trends and thus did not fundamentally alter the conclusions drawn.

Analysis of Categorical Values:

- An analysis of the top and bottom 10 occurrences within the *'track_genre'*, *'artists'*, and *'album_name'* columns was conducted.
- **Most Frequent Occurrences:**
 - For *track_genre*, given the large dataset size (over 100,000 entries), the top 10 most frequent genres, as depicted in the bar chart, all had value counts exceeding 1000. These genres, presented alphabetically, range from "acoustic" to "brazil".
 - For *artists*, the high frequency of certain artists was notably linked to the number of genres associated with their tracks, as evidenced by the difference between "Unique Tracks", which were identified using pandas' `unique()` function to identify the unique track count for the following artists listed on the graph and below, and "Total Tracks". *Refer to the table below.*

Artist	Career Span	Unique Tracks	Total Tracks
<i>The Beatles</i>	1960–1970	149	279
<i>George Jones</i>	1953–2013	215	271
<i>Stevie Wonder</i>	1961–present	16	236
<i>Linkin Park</i>	1996–2017, 2023–present	84	224
<i>Ella Fitzgerald</i>	1934–1993	18	222
<i>Prateek Kuhad</i>	2011–present	46	217
<i>Feid</i>	2013–present	12	202
<i>Chuck Berry</i>	1953–2017	13	190
<i>Håkan Hellström</i>	1991–present	122	183
<i>OneRepublic</i>	2002–present	59	181

Table 1: Most Frequent Artists: Career Spans, Unique Tracks, and Total Tracks

- Initially, it was hypothesized that artists with longer career spans would naturally accumulate more tracks. However, further investigation revealed that the total track count for an artist in this dataset is more strongly correlated with the number of genres associated with their songs, as previously mentioned.
- Interestingly, the most frequently occurring *album_name* entries appeared to be predominantly seasonal albums rather than general releases.
- Visualizing the least frequent counts for *'track_genre'*, *'artists'*, and *'album_name'* proved unhelpful due to the large number of unique entries, with the majority having a frequency of only 1. This longtail distribution made a meaningful "bottom 10" visualization impractical.

Higher Level Analysis:

1. Top 10 Genres by Average Popularity:

A key question to address was identifying the top 10 most "popular" genres within this dataset. The exploratory analysis revealed a substantial number of data entries for various genres. For artists, the number of *unique tracks* for the most frequent artists was comparatively lower than their total track count due to multi genre classifications.

Utilizing a pie chart to illustrate the top 10 genres by average popularity as percentages, it is evident that *pop-film* is the most popular genre, followed by *k-pop*, and so forth.

Percentages of Top 10 Genres by Average Popularity

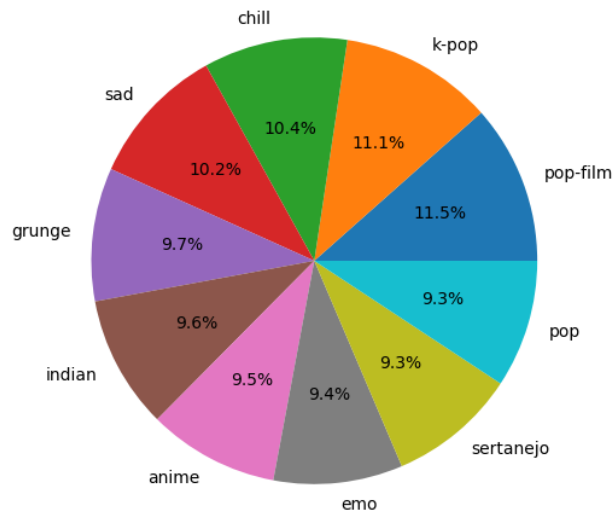


Figure 1: Top 10 Genres by Average Popularity

2. Top 10 Artists by Average Popularity:

Following the analysis of genres, identifying the artists with the highest average popularity was another important aspect to consider.

Again, employing a pie chart to visualize percentage values, the following artists emerged as the top 10 in terms of average popularity.

Percentages of Top 10 Artists by Average Popularity

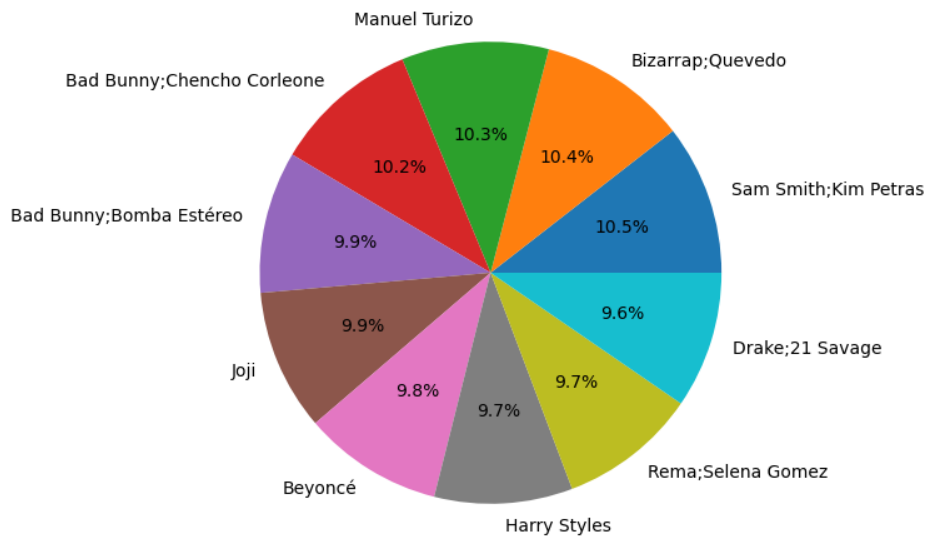


Figure 2: Top 10 Artists by Average Popularity

The table below shows the top 10 artists based on their average popularity scores, from the chart above. This was accomplished by using pandas groupby() function to group tracks by artist and calculate the mean popularity for each one. The nlargest(10) method was then used to extract the top 10 most popular artists, which were also visualized using a pie chart to emphasize their percentages in the dataset. As a result, the genres associated with these artists were identified by filtering the dataset by artist name and using the unique() function to return the distinct genres linked to each artist. Due to this, conclusions were drawn of not only the top 10 artist(s), but what genres they belonged in, which also could be used to ask if there were any similarities from the previous pie chart visualization from the first analysis.

Artist(s)	Genre(s)
<i>Sam Smith; Kim Petras</i>	Dance & Pop
<i>Bizarrap; Quevedo</i>	Hip-Hop
<i>Manuel Turizo</i>	Latin, Latino, reggae, reggaeton
<i>Bad Bunny; Chencho Corleone</i>	Latin, Latino, reggae, reggaeton
<i>Bad Bunny; Bomba Estéreo</i>	Latin, Latino, reggae, reggaeton
<i>Joji</i>	Pop
<i>Beyoncé</i>	Dance
<i>Harry Styles</i>	Pop
<i>Rema; Selena Gomez</i>	Pop
<i>Drake; 21 Savage</i>	Hip-Hop

Table 2: Top 10 Artists Genres

3. Linear Regression: Acousticness vs Energy:

Note: Duplicate songs were dropped

Null Hypothesis

These two categories are uncorrelated.

Alternative Hypothesis

The correlation among the variables are nonzero.

The slope of between the variables is a negative trend of **-0.96**, ($r = -0.73$ $p = 0.0$). However as seen in the heatmap before, this is a strong negative correlation. But the P-Value is statistically significant as it can have impact on a song. Discovering this, the null hypothesis can be rejected.

To describe the relationship between *acousticness* and *energy*, A definition of both of these qualities is important before we can go further. Energy in a song can typically be defined as "excitement" or "intensity" that a listener can feel, whereas acousticness can be defined as a instrument not being amplified through electronic means. Using these two definitions, acoustic songs are generally mellow, which energy is not as present all the time in an acoustic song compared to a rock, hip-hop, or any song that is "aggressive". Which makes sense given the regression chart because acoustic songs do not fit the description of what energy is in a song.

To reiterate, the linear regression showcases a strong negative correlation between these two features, as energy increases, acousticness decreases.

4. K-Means Clustering:

Using K-Means Clustering, an unsupervised machine learning technique, made it possible to answer the question of what defines a music genre. The following results were interpreted in the table below with more specifics for each cluster below the table.

cluster	0	1	2	3	4	5	6	7	8
danceability	0.530549	0.670655	0.347784	0.573955	0.472004	0.470193	0.586176	0.522175	0.716733
energy	0.738668	0.507735	0.169670	0.670453	0.815388	0.313206	0.744782	0.758014	0.768906
loudness	-6.119909	-9.221327	-21.373121	-11.163560	-5.236839	-11.710334	-8.409667	-6.984827	-5.797511
speechiness	0.067194	0.081390	0.050886	0.836033	0.103635	0.044713	0.070921	0.085794	0.092800
acousticness	0.102983	0.588553	0.863385	0.732562	0.111268	0.747609	0.104530	0.281898	0.137521
instrumentalness	0.028417	0.034802	0.827048	0.009980	0.036198	0.031206	0.797010	0.068894	0.021262
liveness	0.177058	0.164539	0.161280	0.652586	0.196743	0.159251	0.170111	0.753127	0.167646
valence	0.316666	0.657978	0.185566	0.442053	0.477254	0.298959	0.332750	0.509049	0.725386
tempo	109.941132	117.154248	103.203736	101.238308	164.595194	112.773173	126.815542	123.602399	116.373098
explicit	0.109307	0.075472	0.002445	0.560550	0.110468	0.021264	0.036178	0.057371	0.126732

Figure 3: K-Means Cluster Results

Cluster	Interpreted Genre
Cluster 0	Electronic Dance Music
Cluster 1	Chill or Mellow
Cluster 2	Instrumental Acoustic
Cluster 3	Spoken Word or Hip-Hop
Cluster 4	High-Energy Pop, Electronic Dance, or Rock-Pop
Cluster 5	Soft Acoustic
Cluster 6	Electronic Instrumental
Cluster 7	Live Performance/Energetic Recording
Cluster 8	Club Hits/Feel-Good Dance

Table 3: Interpreted Music Genres by Cluster

Cluster 0:

- Moderate Danceability (0.53)
- High Energy (0.73)
- Decent Valence (0.31)
- Low Acousticness (0.10)
- Very Low Instrumentalness (0.02)
- **Interpretation:** Likely Electronic Dance Music due to its high energy, moderate danceability, and low acoustic/instrumental presence.

Cluster 1:

- Moderate Danceability (0.67)
- Moderate Energy (0.50)
- Low Loudness (-9.22)
- Moderate Acousticness (0.58)
- **Interpretation:** Suggests Chill or Mellow music characterized by moderate danceability and acousticness, and lower energy/loudness.

Cluster 2:

- High Acousticness (0.86)

- High Instrumentalness (0.82)
- Very Low Loudness (-21)
- Low Energy (0.16)
- **Interpretation:** Points to Instrumental Acoustic music, evident from the high acousticness and instrumentalness coupled with low energy and loudness.

Cluster 3:

- Moderate Danceability (0.57)
- Moderate-High Energy (0.67)
- Medium Loudness (-11.16)
- High Speechiness (0.84)
- **Interpretation:** Indicates Spoken Word or Hip-Hop, defined by high speechiness and moderate danceability/energy.

Cluster 4:

- High Energy (0.82)
- Fairly High Loudness (-5.24)
- Moderate Danceability (0.47)
- Low Acousticness (0.11)
- Low Instrumentalness (0.03)
- **Interpretation:** Could be High-Energy Pop, Electronic Dance, or Rock-Pop, given the high energy/loudness and low acoustic/instrumental qualities.

Cluster 5:

- Low Danceability (0.47)
- Low Energy (0.31)
- Medium Loudness (-11.71)
- High Acousticness (0.75)
- **Interpretation:** Likely Soft Acoustic music, characterized by low danceability and energy alongside high acousticness.

Cluster 6:

- High Energy (0.74)
- Moderate Danceability (0.59)
- Mid-Range Loudness (-8.41)
- High Instrumentalness (0.80)
- **Interpretation:** Suggests Electronic Instrumental music due to its high energy and instrumentalness with moderate danceability.

Cluster 7:

- Moderate Danceability (0.52)
- High Energy (0.76)
- Mid-Range Loudness (-6.98)
- High Liveness (0.75)
- **Interpretation:** Indicates Live Performances or energetic recording sessions, marked by high energy and liveness with moderate danceability.

Cluster 8:

- Highest Danceability (0.72)
- Highest Energy (0.77)
- High Loudness (-5.80)
- Low Acousticness (0.14)
- High Valence (0.73)
- **Interpretation:** Points to Club Hits or feel-good dance tracks, characterized by high danceability, energy, loudness, and valence, with low acousticness.

5. Explicit vs Non-Explicit: Popularity Comparison:

Note: Duplicate songs were dropped

By conducting a T-Test, seeing the difference between explicit and non-explicit popular songs will show if explicit words matter.

Null Hypothesis:

The means of these two groups are the same.

Alternative Hypothesis:

The means of these two groups are different.

First evaluating if there are more popular explicit songs on average with non-explicit was conducted. For explicit, a mean of **36.88** was produced and for non-explicit a mean of **32.85** was produced. Looking at the averages at face value, they difference is relatively close, to verify this assertion the T-Test was conducted.

Following the T-test, the statistic value is **14.89**, suggesting a big difference. A P-Value less than 0.01 ($7.391205139975994 \times 10^{-61}$). After this, the null hypothesis can be rejected as explicit songs tend to be more popular than non-explicit songs.

6. Linear Regression: Loudness vs Energy:

Note: Duplicate songs were dropped

Null Hypothesis

These two categories are not correlated.

Alternative Hypothesis

The correlation among the variables are nonzero.

Because duplicate values were dropped for this linear regression to prevent data skewness. The results of the linear regress are as follows: a moderate positive slope of **15.43**, ($r=0.76$ $p = 0.0$). Which means we can reject the null hypothesis as these two are related and the P-Value showcases significance. Interestingly enough on the graph, as similar to the exploratory analysis for this comparison, there is a massive trend increase for these two categories even with duplicate values in the exploratory analysis.

The relationship between these two are strongly correlated as mentioned. But how so? Energy was previously defined as excitement or intensity in a track. Loudness from reading the word can be interpreted how loud a song is. These two definitions are synonymous with each other which explains the strong positive correlation presented on the graph. As an example a song from the rock genre provides immense energy and loudness, which is an example that expresses a relationship between these two categories. Not only rock, but other genres such as hip-hop or electronic can provide such a correlation between the two.

To reiterate, as energy increases, loudness also increases as these two categories are synonymous with each other which makes sense of why they are positively correlated with each other from the heatmap and through statistical testing.

Conclusion:

After thorough investigation, several findings have been uncovered:

- Firstly, while comparing pie charts for “Top 10 genres among popularity” and “Top 10 artists among popularity”, different genres were listed in each chart, which can be a reflection of a listener’s preferences.
- Moreover, both linear regression charts portray correlations more effectively than in the heatmap and how statistically significant it is.
- Such as in ‘Loudness vs Energy’, the P-Value showed how significant these two variables are alongside looking at a linear regression graph and the exploratory analysis graph.
- Similarly in ‘Energy vs Acousticness’, the graph showed a decrease in between the two variables and the P-Value showcased how significant these values are.
- The clustering analysis revealed distinct song genres and what “creates” that song genre through audio features and other variables.
- Finally the explicit vs non-explicit relationship portrayed preferences among listeners for a song’s popularity. The findings revealed explicit songs were more popular than that of non-explicit.

All in all, this dataset illustrates the ability and how profound data analysis can be not just for music trends but for any dataset. Allowing one to define variables to interpret their meanings and discover correlations between the variables created, it allowed understanding for a listener’s preferences to help create recommendation systems, production, and more.